# CHEMISTRY

**Q1** Consider the following reaction

$S(s) + \frac{3}{2} O_2(g) \rightarrow SO_3(g) + 2x$ KJ

$SO_2(g) + \frac{1}{2} O_2(g) \rightarrow SO_3(g) + y$ KJ

Calculate $\Delta H_r$ for following reaction (in KJ).

$S(s) + O_2(g) \rightarrow SO_3(g)$

(1) $-(x + y)$ (2) $-(2x + y)$ (3) $\frac{x}{y}$ (4) $y - 2x$

**Ans.** (4)

**Sol.** $S(s) + \frac{3}{2} O_2(g) \rightarrow SO_3(g) + 2x$ KJ .....(i)

By eq. (i)

$\Delta H_1 = -2x$ KJ

$SO_2(g) + \frac{1}{2} O_2(g) \rightarrow SO_3(g) + y$ KJ ..... (ii)

By eq. 1 – 2

$\Delta H_3 = -2x - (-y)$

$= (y - 2x)$

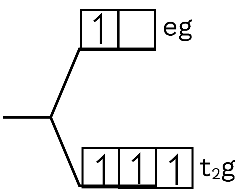
**Q2** The condition and consequences that favour the $t_{2g}^3 e_g^1$ configuration in metal complex are

(1) WFL, high spin complex (2) WFL, low spin complex

(3) SFL, high spin complex (4) SFL, low spin complex

**Ans.** (1)

**Sol.** $t_{2g}^3 e_g^1$




WFL and high spin complex.

**Q3** When ethane 1,2-diamine is progressively added to aqueous solution of Ni(II) chloride the sequence of the colour change observed will be:

(1) Violet → Blue → Pale blue → Green

(2) Pale blue → Blue → Green → Violet

(3) Green → Pale blue → Blue → Violet

(4) Pale blue → Blue → Violet → Green

**Ans.** (3)

**Sol.** $\underset{\text{Green}}{[Ni(H_2O)_6]^{+2}} \longrightarrow \underset{\text{Pale Blue}}{[Ni(H_2O)_4(en)]^{2+}} \longrightarrow \underset{\text{Blue}}{[Ni(H_2O)_2(en)_2]^{2+}} \longrightarrow \underset{\text{Violet}}{[Ni(en)_3]^{2+}}$

**Q4** Given below are two statements:

**Statement-I:** First Ionisation energy of Ge is greater than Si.

**Statement-II:** First Ionisation energy of Pb is greater than Sn

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

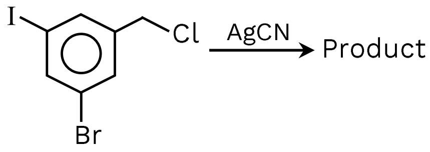
(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

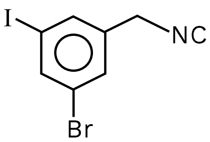
(4) Statement I is incorrect but Statement II is correct

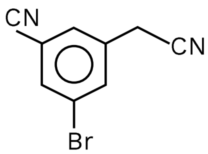
**Ans.** (4)

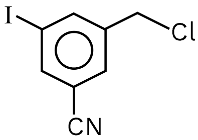
**Q5**

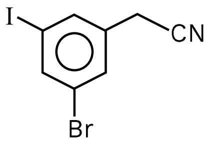



What will be final product?

(1) 

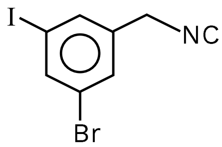

(2) 


(3) 

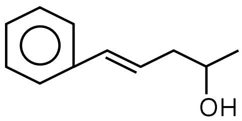

(4) 


**Ans.** (1)

**Sol.**

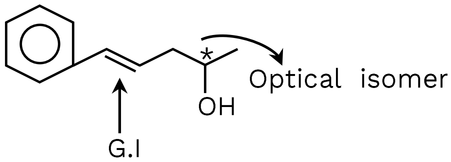



**Q6** Number of stereoisomers for given compound?




**Ans.** (4)

**Sol.**




**Q7** Match the List:

| **List-I (Cations)** | **List-II (B.M. Values)** |
|---|---|
| (P) $Sc^{+3}$ | (1) 2.82 |
| (Q) $V^{2+}$ | (2) 0.0 |
| (R) $Ti^{3+}$ | (3) 3.87 |
| (S) $Ni^{2+}$ | (4) 1.73 |

(1) P→3; Q→2; R→4; S→1

(2) P→2; Q→3; R→4; S→1

(3) P→1; Q→3; R→4; S→2

(4) P→2; Q→4; R→3; S→1

**Ans.** (2)

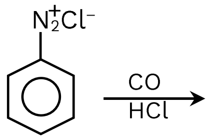
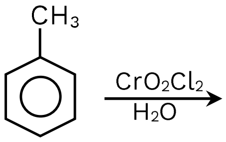
**Sol.** $Sc^{+3} \rightarrow 3d^0 \rightarrow n = 0 \qquad \mu = 0$

$V^{2+} \rightarrow 3d^3 \rightarrow n = 3$ $\mu = 3.87$

$Ti^{3+} \rightarrow 3d^2 \rightarrow n = 1$ $\mu = 1.73$

$Ni^{2+} \rightarrow 3d^8 \rightarrow n = 2$ $\mu = 2.83$

**Q8** Match the following

| **List -I** | **List -II** |
|---|---|
| **(Name)** | **(Reaction)** |
| (A) Gettermann reaction | (P) $R-\overset{O}{\overset{\parallel}{C}}-Cl \xrightarrow[BaSO_4]{H_2/Pd}$ |
| (B) Stephan's reaction | (Q)  |
| (C) Rosenmund reaction | (R) $R-C\equiv N \xrightarrow[HCl]{SnCl_2}$ |
| (D) Etard Reaction | (S)  |

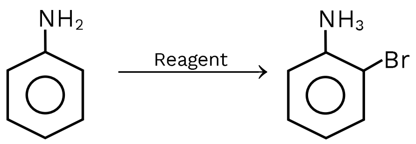
(1) A→Q, B→R, C→P, D→S (2) A→R, B→P, C→Q, D→S

(1) A→Q, B→R, C→P, D→S (2) A→R, B→P, C→Q, D→S

(3) A→Q, B→P, C→R, D→S (4) A→Q, B→R, C→S, D→P

**Ans.** (1)

**Sol.** Conceptual

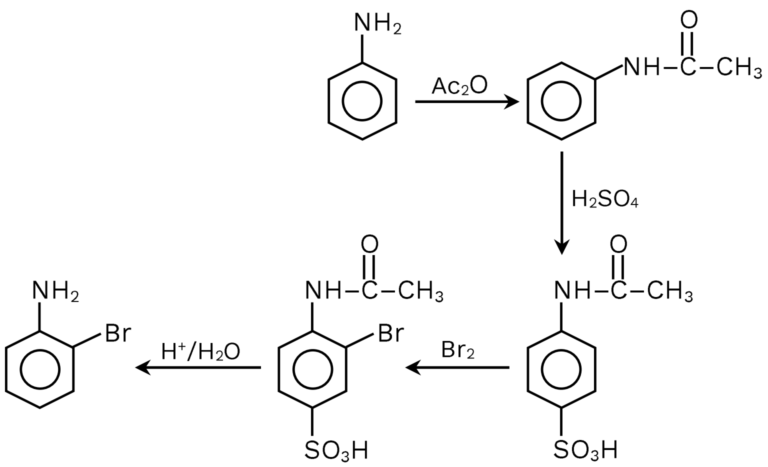
**Q9** 

Suitable reagent is:

(1) $Ac_2O$, $H_2SO_4$, $H_2O$, NaOH (2) $Ac_2O$, $H_2SO_4$, $Br_2$, NaOH

(3) $Ac_2O$, $Br_2$, $H_2O$, $H_2SO_4$, NaOH (4) $Ac_2O$, $H_2SO_4$, $H_2O$, NaOH

**Ans.** (2)

**Sol.** 

**Q10** In a compound contains 54.2% carbon, 9.2% of hydrogen and rest are oxygen. What is molecular formula of compound, if molecular mass is 132 g/mol?

(1) $C_6H_{12}O_3$ (2) $C_4H_{12}O_3$ (3) $C_4H_{12}O_6$ (4) $C_6H_{13}O_6$

**Ans.** (1)

**Sol.**

| C | H | O |
|---|---|---|
| $\frac{54.2}{12}$ | $\frac{9.2}{1}$ | $\frac{36.6}{16}$ |

$100 - (54.2 - 9.2) = 36.6$

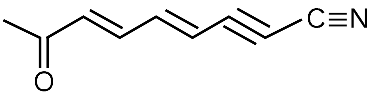
Divide by $\frac{36.6}{16}$ to all

Hence $(C_2H_4O)_n$

$n = \frac{132}{44} = 3$

$C_6H_{12}O_3$

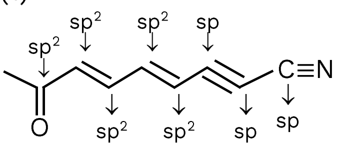
**Q11** Number of sp–sp² Hybridized Carbon atom ?



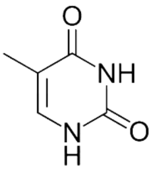
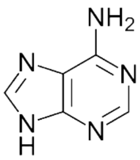
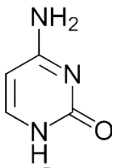
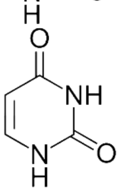

(1) 3, 5 (2) 3, 6 (3) 2, 6 (4) 4, 5

**Ans.** (1)

**Sol.**




**Q12** Match the following column

| **List -I** | **List -II** |
|---|---|
| **Name** | **Structure** |
| (A) Adenine | (P)  |
| (B) Cytosine | (Q)  |
| (C) Uracil | (R)  |
| (D) Thymine | (S)  |

(1) A→Q, B→R, C→P, D→S (2) A→Q, B→R, C→S, D→P

(3) A→Q, B→P, C→R, D→S (4) A→R, B→Q, C→S, D→P

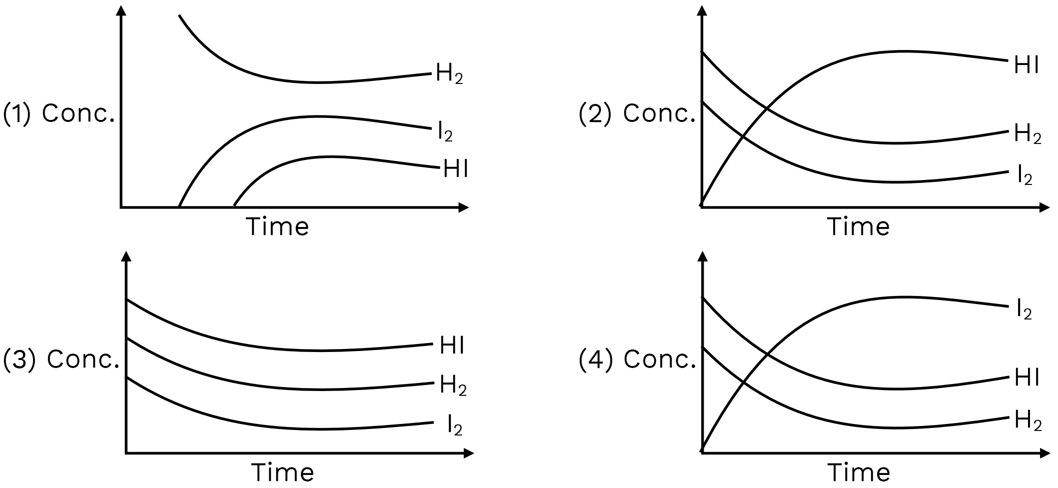
**Ans.** (2)

**Sol.** NCERT Diagram Based

**Q13** Consider the following gaseous reaction

$H_2(g) + I_2(g) \rightarrow 2HI$

The above reactionis started wit 'a' moles of $H_2$ and 'b' moles of $I_2$ in a closed to container at a certain temperature T(K) till the equilibrium is established. Wic one of the following plots correctly describes the progress of reaction?




**Ans.** (2)

**Sol.** The initial concentration of product (HI) should be zero.

**Q14** A Hydrocarbon X which has molar mass 80 g Contains 90% carbon. Find degree of unsaturation in X

**Ans.** (3)

**Sol.** $m_C = 80 \times \frac{90}{100} = 72g; n_C = \frac{72}{12} = 6$

$m_H = 80 \times \frac{10}{100} = 8g; n_C = \frac{8}{1} = 8$

$X = C_6H_8$

Degree of unsaturation = 3

**Q15** For $MX_2$, the observed molar mass: 65.6

Normal molar mass: 165

Find percentage dissociation.

**Ans.** (75)

**Sol.** $MX_2 \longrightarrow M^{2+} + 2X^-$

$i = 1 + (n - 1)\alpha$

$2.5 = 1 + (3 + 1)\alpha$

$\alpha\% = 0.75$

$\alpha\% = 75\%$

**Q16** 1M acid is reacted with 1M base. In which case temperature increase will be high.

(1) 35 ml HCl + 25 ml NaOH

(2) 35 ml NaOH + 25 ml $CH_3COOH$

(3) 50 ml HCl + 10 ml NaOH

(4) 30 ml HCl + 30 ml NaOH

**Ans.** (4)

**Sol.** 30m mole

$\because$ SA + SB

Release high energy which cause temperature increase.

**Q17** For a reaction

$$K = \sqrt{\frac{K_1 \times K_3}{K_2}}$$

Find $(E_a)$

If $E_{a1}$ = 10 kJ; $E_{a2}$ = 30 kJ ;$E_{a3}$ = 60 kJ

(1) 30 (2) 25 (3) 15 (4) 20

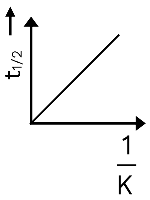
**Ans.** (4)

**Sol.** $\because K = Ae^{-\frac{E_a}{RT}}$

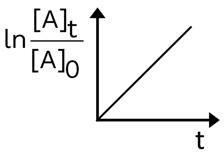
Put the value of K in $K = \sqrt{\frac{K_1 \times K_3}{K_2}}$

$$Ae^{-\frac{(E_a)_{eff}}{RT}} = \sqrt{\frac{A_1e^{-\frac{E_a}{RT}} \times A_3e^{-\frac{E_a}{RT}}}{A_2e^{-\frac{E_a}{RT}}}}$$

**Q18** Given below are two statements :

**Statement-I:** 

order of reaction is 1.

**Statement-II:** 

order of reaction is 1.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

**Ans.** (2)

**Sol.** Statement-I is $t_{1/2} = \frac{\ln 2}{K}$

$y = mx$

Statement-II is

$\because A_t = Ae^{-kt}$

$$\frac{A_t}{A_0} = e^{-Kt}$$

$$\ln\frac{A_t}{A_0} = -Kt$$

$y = -mx$

**Q19** Which one is act as strong reducing agent.

(a) $E^{o}_{MnO_4^-/Mn^{2+}} = +1.71$ (b) $E^{o}_{Cl/Cl^-} = +1.36$

(c) $E^{o}_{Cr^{3+}/Cr} = -1.31$ (d) $E^{o}_{Cr_2O_7^{2-}/Cr^{3+}} = -1.38$

(1) $Mn^{2+}$ (2) $MnO_4^-$ (3) Cr (4) $Cr^-$

**Ans.** (3)

**Sol.** Reducing agent $\Rightarrow$ high oxidatin potential.

$Cr^{3+}$ act as strongest reducing agent